Transport of the deficit-four horizon along fixed-seed returns

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Let

$$x_{t+1} = U_+(x_t) := \frac{3x_t + 1}{2^{a_t}}, \qquad a_t := v_2(3x_t + 1), \qquad A_t := \sum_{i=0}^{t-1} a_i, \tag{1}$$

where every x_t is a positive odd integer. We assume that every prefix from x_0 is a strict barrier:

$$2^{A_t} < 3^t \qquad (t \geq 1). \tag{2}$$

Lemma 1 (exact return lift and extension law)**.** *Suppose that the length-L exponent prefix occurs again at position $d > 0$:*

$$(a_d, \ldots, a_{d+L-1}) = (a_0, \ldots, a_{L-1}). \tag{3}$$

Then there is an integer $t_L \geq 1$ such that

$$\boxed{x_d = x_0 + t_L 2^{A_L+1}, \qquad x_{d+L} = x_L + 2t_L 3^L.} \tag{4}$$

For $h \geq 0$, put

$$B_h := A_{L+h} - A_L. \tag{5}$$

The occurrence in (3) extends through the next h exponent letters if and only if

$$\boxed{2^{B_h} \mid t_L.} \tag{6}$$

Whenever (6) holds,

$$\boxed{x_{d+L+h} = x_{L+h} + 2\frac{t_L}{2^{B_h}} 3^{L+h}.} \tag{7}$$

In particular,

$$\min\{v_3(x_{d+L+h} + 1), L + h\} = \min\{v_3(x_{L+h} + 1), L + h\}. \tag{8}$$

Proof. A positive exponent word of total exponent A determines one odd seed class modulo 2^{A+1}. We include the standard induction. Suppose a prefix of length k and total exponent A has representative ρ. Replacing its initial value by $\rho + \ell 2^{A+1}$ replaces its endpoint by

$$x_k(\rho) + 2\ell 3^k.$$

Appending an exponent a requires

$$3x_k(\rho) + 1 + 6\ell 3^k \equiv 2^a \pmod{2^{a+1}}.$$

After division by 2, the coefficient of ℓ is odd, so this congruence selects one class $\ell \pmod{2^a}$. This proves the cylinder assertion.

The two starting states in (3) therefore agree modulo 2^{A_L+1}, which gives the first identity in (4). Condition (2) and the positive affine carry give $x_d > x_0$, so $t_L \geq 1$. Applying the common length-L affine map to the starting-state difference gives the second identity.

The occurrence extends through h further letters exactly when x_d and x_0 lie in the same length-$(L+h)$ cylinder. By the first identity in (4), this is equivalent to

$$2^{A_{L+h}+1} \mid t_L 2^{A_L+1},$$

which is (6). Applying the common length-$(L+h)$ affine map proves (7), and reduction of (7) modulo 3^{L+h} proves (8). □

Let

$$\mathcal{S}_4 := \{\varnothing, 12, 112, 1112, 1113, 1122, 1212, 11113\} \tag{9}$$

be the proved deficit-four stem set. A stem s matches x_0 if it is the initial exponent word of x_0. For a matching stem of length h, put

$$\nu_s := v_2(x_h + 1). \tag{10}$$

The proved fixed-start horizon is

$$R_4(x_0) := \max_{\substack{s \in \mathcal{S}_4 \\ s \text{ matches } x_0}} (|s| + \nu_s - 1). \tag{11}$$

Define also the finite transport threshold

$$\Gamma_4(x_0) := \max_{\substack{s \in \mathcal{S}_4 \\ s \text{ matches } x_0}} (A_{|s|} + \nu_s). \tag{12}$$

Theorem 2 (fixed-seed horizon transport)**.** *Assume (3), $L \geq 5$, and*

$$A_L + 1 > \Gamma_4(x_0). \tag{13}$$

Then

$$\boxed{R_4(x_d) = R_4(x_0).} \tag{14}$$

Since $A_L \geq L$, condition (13) holds for every sufficiently long returned prefix of the fixed seed x_0.

Proof. Every stem in $\mathcal{S}_4$ has length at most 5. Thus (3) and $L \geq 5$ show that exactly the same stems match x_d and x_0. Fix one such stem s and put $h := |s|$. Applying its common affine map to the first identity in (4) gives

$$x_{d+h} - x_h = t_L 3^h 2^{A_L+1-A_h}. \tag{15}$$

By (12) and (13),

$$A_L + 1 - A_h > \nu_s = v_2(x_h + 1).$$

Equation (15) therefore implies

$$v_2(x_{d+h} + 1) = v_2(x_h + 1) = \nu_s. \tag{16}$$

Every term in the maximum defining R_4 is the same at the two starts, so (14) follows. □

Corollary 3 (returned-start exclusion)**.** *Under the hypotheses of Theorem 2, suppose also that*

$$L > R_4(x_0). \tag{17}$$

Let $r \geq L$. If the length-r exponent block beginning at x_d is a strict barrier block, then

$$\boxed{v_3(x_{d+r} + 1) \leq r - 5.} \tag{18}$$

In particular, if x_d begins an infinite strict-barrier ray, then no length $r \geq L$ from this returned start can be a deficit-four synchronization hit. Any such hit obtained from the return construction must change the block start.

Proof. Theorem 2 gives $R_4(x_d) = R_4(x_0) < L \leq r$. The fixed-start deficit-four horizon theorem at x_d now gives (18). The remaining assertions follow immediately. □

Theorem 4 (first-branch exponent law)**.** *Assume the hypotheses of Theorem 2. Let h be a finite maximal common extension of the returned prefix, and put*

$$N := L + h, \qquad B := A_N - A_L, \qquad T := \frac{t_L}{2^B}. \tag{19}$$

Let

$$y := x_N, \qquad y' := x_{d+N} = y + 2T3^N, \tag{20}$$

and denote their distinct next exponents by

$$a := v_2(3y+1), \qquad a' := v_2(3y'+1). \tag{21}$$

Put $c := v_2(T)$ and $z := (3y+1)/2^a$. Then $c < a$, and the first branch is governed exactly by

$$\boxed{\begin{aligned} c+1 < a \quad &\Longrightarrow \quad a' = c+1, \\ c+1 = a \quad &\Longrightarrow \quad a' = a + v_2\left(z + \frac{T}{2^{a-1}}3^{N+1}\right) > a. \end{aligned}} \tag{22}$$

Proof. Maximality of h and the extension law (6) give

$$2^B \mid t_L, \qquad 2^{B+a} \nmid t_L,$$

so $c < a$. Equation (20) gives

$$3y' + 1 = 2^a z + 2T3^{N+1}. \tag{23}$$

The two terms on the right have 2-adic valuations a and $c+1$. If they are distinct, their sum has the smaller valuation. If they are equal, factoring out 2^a gives the second line of (22); the two remaining summands are odd, so the displayed valuation is positive. □